

1 Solution — GIAM 9.3

Problem 1

1.1 Problem

There is a scenario where the proof we have sketched for Monge’s circle theorem doesn’t really work. Can you envision it? *Hint: consider two relatively large spheres and one that is quite small.*

1.2 The Sketched Proof — and Its Fragile Step

Recall the lift-to-3D argument:

1. Embed the plane of the three circles as $z = 0$ in $\mathbb{R}^3$.
2. Replace each circle by a sphere of the same radius and centre.
3. For each *pair* of spheres, the tangent **cone** has its apex *exactly* at the external-tangent intersection of the two original circles in $z = 0$.
4. **Key step.** “There is a plane that touches all three spheres tangentially from above.”
5. That upper tangent plane meets $z = 0$ in a line.
6. Each tangent cone is tangent to its two spheres *and* to the upper plane, so the cone’s apex lies on the upper plane. Combined with $z = 0$, the three apexes lie on the line of intersection. ■

Steps 1, 2, 3, 5, 6 are routine. **Step 4 is the gamble** — and it is exactly where the proof can collapse.

1.3 When Step 4 Fails — Two Large Spheres, One Small

Picture two beach balls on a table with a small marble between them. The plane resting on top of both beach balls is *high overhead*. The marble’s top is barely off the table — the plane sails right over the marble without ever touching it.

There is no single plane tangent to *all three* spheres from above: any plane that touches the two big ones is way above where the small one reaches.

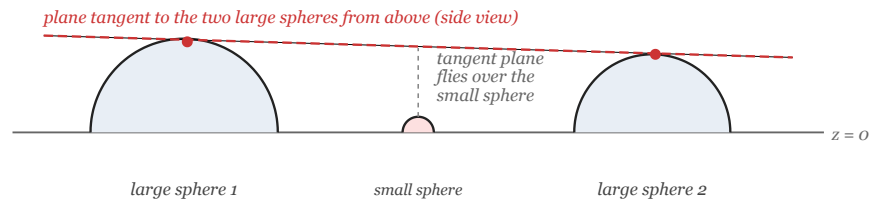


Figure 1.1: Side view. The dashed red line is the cross-section of the plane tangent to the two large spheres from above. It passes far above the small sphere — there is no plane tangent to all three from above.

1.4 Concrete Example with Numbers

Place the three spheres on $z = 0$ with centres and radii:

Sphere	Centre	Radius
1	(0, 0, 0)	6
2	(30, 0, 0)	5
3	(15, 4, 0)	1

Table 1.1

No circle is inside another:

- $|C_1 - C_3| = \sqrt{225 + 16} \approx 15.52 > |R_1 - R_3| = 5. \checkmark$
- $|C_2 - C_3| \approx 15.52 > 4. \checkmark$
- $|C_1 - C_2| = 30 > 1. \checkmark$

Suppose an upper tangent plane existed, with unit normal $\mathbf{n} = (n_x, n_y, n_z)$, $n_z > 0$. Tangency from above to sphere i means the plane sits at height $\mathbf{n} \cdot \mathbf{C}_i + R_i$; equating across pairs:

$$\mathbf{n} \cdot (\mathbf{C}_1 - \mathbf{C}_2) = R_2 - R_1, \quad \mathbf{n} \cdot (\mathbf{C}_1 - \mathbf{C}_3) = R_3 - R_1.$$

Substituting:

$$\mathbf{n} \cdot (-30, 0, 0) = -1 \implies n_x = \frac{1}{30},$$

$$\mathbf{n} \cdot (-15, -4, 0) = -5 \implies -\frac{1}{2} - 4n_y = -5 \implies n_y = \frac{9}{8} = 1.125.$$

But $\mathbf{n}$ must satisfy $n_x^2 + n_y^2 + n_z^2 = 1$. Already

$$n_x^2 + n_y^2 = \frac{1}{900} + \frac{81}{64} \approx 1.267 > 1,$$

so no real n_z exists. **No upper tangent plane to all three spheres exists.** Step 4 of the proof has no leg to stand on.

1.5 Why the *Smallness* of Sphere 3 is the Culprit

The constraint that broke the system was

$$\mathbf{n} \cdot (\mathbf{C}_1 - \mathbf{C}_3) = R_3 - R_1.$$

When R_3 is small and R_1 is large, the right-hand side is roughly $-R_1$ — a *big* number. The left-hand side is bounded by $|\mathbf{C}_1 - \mathbf{C}_3|$ (Cauchy-Schwarz with $|\mathbf{n}| \leq 1$).

So a plane can exist only if

$$|\mathbf{C}_1 - \mathbf{C}_3| \geq |R_3 - R_1| \approx R_1.$$

The non-containment condition only requires $|\mathbf{C}_1 - \mathbf{C}_3| \geq |R_1 - R_3|$ — *the same lower bound*. So existence of the upper tangent plane is barely compatible with non-containment, and the *combined* constraint from the second pair usually pushes the required $\mathbf{n}$ over the unit-vector limit.

In the example above, increasing R_3 from 1 to about 2 would push n_y down to 1 (the borderline). Below that threshold, no plane.

1.6 The Theorem Still Holds

Monge’s theorem itself is *not* in jeopardy. The three external-tangent intersection points are still collinear in the failure scenario — what fails is the *particular proof we sketched*, not the conclusion.

The classical proof uses **similitude**: each external-tangent intersection is the *external centre of similitude* of a pair of circles, i.e. the point that scales one circle into the other by a positive factor. A short composition-of-homotheties argument shows that the three external centres of similitude of any three circles are collinear — regardless of radii or position, no 3D lift needed.

So the textbook’s failure is a failure of the *illustration*, not the *theorem*.

1.7 Remark — The “Equal Radii” Degeneracy is a Different Failure

The textbook flagged a separate degeneracy: when *all three* spheres have the same radius, the tangent cones become *cylinders* (apex at infinity) and the upper/lower tangent planes become parallel — meeting in a line “at infinity”. The collinearity statement becomes a projective one.

The small-sphere failure is *not* this. There, the radii are genuinely different, all three apexes are finite real points, and Monge’s theorem is non-trivial — but the *3D-lift proof* misfires because the would-be upper tangent plane doesn’t exist as a real plane.

1.8 Remark — A Quantitative Threshold

Boiling the algebra down, the upper tangent plane to all three spheres exists iff the unique (n_x, n_y) satisfying

$$\mathbf{n} \cdot (C_1 - C_2) = R_2 - R_1, \quad \mathbf{n} \cdot (C_1 - C_3) = R_3 - R_1$$

obeys $n_x^2 + n_y^2 \leq 1$.

Geometrically this is a condition on how the radii relate to the *triangle* formed by the centres. With two large equal-ish radii and a small third radius, the constraint from the

small sphere demands a near-horizontal $\mathbf{n}$ — and combined with the constraint from the large pair, the total horizontal magnitude tips over 1.

The borderline $n_x^2 + n_y^2 = 1$ corresponds to a *vertical* tangent plane (i.e. $n_z = 0$), a degenerate case where the “plane above all three” is actually edge-on.

1.9 Remark — Why “Above” Versus “Below” Doesn’t Save Us

One might hope that *some* tangent plane to all three spheres exists, even if not the “above all” one — e.g. one that touches the big ones from above and the small one from below. In our concrete example, *all eight* sign-combinations of above/below fail:

sphere 1	sphere 2	sphere 3	required n_x from spheres 1-2	required n_y from spheres 1-3
above	above	above	$\frac{1}{30}$	$\frac{9}{8}$
above	above	below	$\frac{1}{30}$	$\frac{11}{8}$
above	below	above	$-\frac{11}{30}$	varies
above	below	below	$-\frac{11}{30}$	varies
below	above	above	$\frac{11}{30}$	varies
below	above	below	$\frac{11}{30}$	varies
below	below	above	$-\frac{1}{30}$	$-\frac{11}{8}$
below	below	below	$-\frac{1}{30}$	$-\frac{9}{8}$

Table 1.2

In each row the required n_y (or n_x in the asymmetric rows) ends up exceeding 1 in magnitude. *No* tangent plane to all three spheres exists in this configuration — not even a “mixed-sign” one.

So the 3D-lift proof is genuinely unsalvageable for this configuration: the similitude argument is the way out.

1.10 Remark — A Pithy Description

Two beach balls on a table with a marble between them.

The plane resting on top of both beach balls is high overhead. Lower it — it scrapes the beach balls but flies over the marble. Tilt it — one beach ball pushes back up. *Any* plane that touches all three would have to dip down to scrape the marble *and* climb up to touch the beach balls, which a flat plane cannot do.

In 2D — looking straight down at the table — the three pairs of external tangent lines do still meet at three points, and those three points *are* collinear. Just not because of any beautifully-resting plane above. The collinearity is real; the picture-book proof is not the right intermediary to see it.

1.11 Remark — A Cleaner Way to Frame the Failure

The 3D-lift proof works whenever a *common tangent plane* to all three spheres exists (above or below or mixed). In *most* configurations, at least one such plane does exist. The failure scenario — extreme size disparity, small sphere tucked close to the others — pushes the configuration into a region of “no common tangent plane at all”.

A way to see this geometrically: every common tangent plane corresponds to a point on the unit sphere of normals satisfying *two* linear equations (one for each pair of spheres). The intersection of the unit sphere with two planes generically gives two points (or one, or zero). When the two planes pass *outside* the unit sphere, there are no real solutions — and that is precisely what extreme size disparity provokes.

So the proof’s “there is a plane” is, more honestly: “there is generically a plane”. The hint pushes us into the *non-generic* corner where the generic existence fails.